

Health Insurance and Entrepreneurship Lock*

Ziyang Jia[†]

September 17, 2026

Abstract

Entrepreneurship lock arises when limited access to health insurance discourages self-employment. I study whether this constraint binds for young adults by exploiting the Affordable Care Act dependent coverage mandate, which extended parental plan eligibility to age 26. In a difference-in-differences design, I find that the mandate raised self-employment by 26%. I then develop a mean-variance framework to separate the subsidy channel (lower expected premiums and out-of-pocket medical costs) from the risk-reduction channel (lower variance of total payments) and find that the risk-reduction channel is the primary mechanism. Finally, I examine distributional effects and find that women and the unhealthy benefit the most.

JEL Codes: I13 · L26 · J23 · J32

Keywords: Health Insurance · Health · Entrepreneurship Lock · Self-employment · ACA

*I am indebted to the guidance and help from my advisor Cameron M. Ellis. I thank Cameron Ellis, Meghan Esson, Glenn Harrison, Richard Peter, Evan Saltzman, and Nicolas Ziebarth as well as audience members of The University of Iowa, The UGA PhD Student Symposium, The Risk Theory Society, and The American Risk and Insurance Association for helpful comments.

[†]Tippie College of Business, University of Iowa (zijia@uiowa.edu)

1 Introduction

In the United States, health insurance often comes with the job. Nearly half of working-age Americans receive health care through an employer, and only 15% of employed workers are uninsured (IPUMS MEPS, 2021). Among the self-employed, nearly one-third lack coverage. High premiums and limited options in the individual market can deter workers from leaving wage jobs, a pattern known as job lock. This constraint can also block new business formation, which the literature labels entrepreneurship lock. The distinction matters for policy: entrepreneurs generate positive spillovers through job creation, innovation, and productivity growth.¹

This paper studies entrepreneurship lock using the ACA dependent coverage mandate, which allowed young adults to remain on a parent’s plan until age 26. I combine MEPS household and insurance files (2006–2014) in a difference-in-differences design comparing 19–25-year-olds to 27–34-year-olds. The mandate raised self-employment by 0.89 pp, a 26% increase relative to the pre-policy treated mean, indicating that binding insurance constraints limit young adults’ entrepreneurial entry. I then show that this increase in entrepreneurship occurs primarily through the risk reduction value of the insurance as opposed to a direct subsidy effect. Finally, I examine heterogeneity in outcomes and find that women and the unhealthy benefit the most.

A large literature documents job lock from employer-sponsored insurance (Madrian, 1994; Gruber and Madrian, 1994; Hamersma and Kim, 2009; Garthwaite et al., 2014). Related work on the gig economy documents transitions from contract work into small-firm entrepreneurship using tax records (Denes et al., 2025). Evidence on entrepreneurship lock from health insurance is more mixed. Prior studies use federal reforms, including Medicare eligibility at age 65 and the Affordable Care Act (Fairlie et al., 2011; Heim and Yang, 2017; Bailey, 2017; Bailey and Dave, 2019; Yörük and Xu, 2019; Blume-Kohout, 2023, 2024), and state reforms, including New Jersey’s individual health coverage program and Massachusetts health reform (DeCicca, 2010; Niu, 2014). Other work uses whether a worker has access to a spouse’s employer-sponsored plan (Holtz-Eakin et al., 1996; Wellington, 2001), sometimes comparing workers with and without that op-

¹See, e.g., Schmitz Jr (1989); Baumol (1990); Murphy et al. (1991); Aghion and Howitt (1992); Acs and Audretsch (1988); Haltiwanger et al. (2013); Glaeser et al. (2015).

tion. Less is known about whether insurance expansions relax entrepreneurship lock by raising disposable income (a subsidy) or by reducing exposure to health expenditure risk (insurance value), and which populations benefit most.

The dependent coverage mandate is a useful setting because it expands access to insurance independently of the young adult's own employer. Before the mandate, most private plans ended dependent coverage at age 19, or at 23 for full-time students (Cantor et al., 2012). Implementation began in late 2010. An eligible 24-year-old with access to a parent's plan could now leave a wage job to start a business without giving up that coverage. MEPS links insurance plans to employment status, annual premiums, out-of-pocket medical spending, and detailed health measures. These data allow me to ask not only whether the mandate changes self-employment, but how that response relates to expected payments and payment risk.

Identification requires that, absent the mandate, self-employment among ages 19–25 would have followed the same trend as among the older comparison group. The event study shows no clear differential pre-policy trend, with larger positive estimates in 2013–2014. The estimated mandate effect on self-employment remains positive across 256 combinations of demographic fixed effects and when I narrow the comparison to 25- and 27-year-olds. A placebo comparison between older, ineligible cohorts yields an estimate near zero, and controlling for spousal employer coverage leaves the main estimate nearly unchanged.

I then ask whether the mandate works mainly through lower expected annual health-care costs or through lower payment risk. I develop a mean-variance model which motivates both a subsidy channel (expected premiums and out-of-pocket payments, summarized as total pay) and a risk-reduction channel (the conditional variance of total pay). The empirical challenge is separating out the impact of each moment. To do so, I use a multi-stage IV procedure. I first instrument realized total pay using mandate-induced variation and interactions with principal components of baseline demographics. I then square the residuals and instrument those squared residuals using the same instruments. Intuitively, I am exploiting that the mandate impacts the first and second moment differently across different demographics. I find that self-employment does not respond statistically to the first moment, but rises when the second moment falls, holding the first moment constant. This pattern suggests that risk reduction, rather than lower expected costs alone, drives the entry response.

Finally, I examine who benefits from expanded dependent coverage. The mandate effect is 5.0 percentage points larger for young adults with diabetes and 3.4 percentage points larger for those with cancer than for individuals without the respective condition. Women experience an increase 2.3 percentage points larger than men. The response is not uniform across health measures. I do not find precise differential effects for hypertension, cardiovascular disease, or general physical and mental health scores. The results suggest that insurance access affects who becomes self-employed as well as the overall self-employment rate.

This study contributes in three ways. First, I add to the literature on entrepreneurial supply by showing that loosening dependence on employer-sponsored insurance raises self-employment among young adults. The closest comparison is Bailey (2017), who studies the same dependent coverage mandate and finds no statistically detectable increase in self-employment among young adults overall, but an increase among those with disabilities. My positive average estimate differs from that result. MEPS also permits an examination of medical payments, insurance preferences, and specific health conditions within the same setting.

Second, I connect entrepreneurship lock to models of occupational choice under risk and to evidence linking risk preferences to self-employment (Kihlstrom and Laffont, 1979; Hvide and Panos, 2014; Skriabikova et al., 2014; Hochmuth et al., 2025). Rather than asking only whether entrepreneurs are less risk-averse than wage workers, I ask whether health expenditure risk keeps people out of self-employment, holding expected payments fixed. Instrumented total pay and the instrumented squared residual, constructed from mandate-induced variation, let me examine both channels in the same policy setting. The relevant risk is not only the uncertainty generated by a business; it also includes the medical payment risk a worker must bear to become self-employed.

Third, I document heterogeneity in the distributional effects of dependent coverage. Prior work examines health-related selection into self-employment and the effects of the ACA on women's self-employment (Rietveld et al., 2015a; Blume-Kohout, 2023). Blume-Kohout (2024) finds that the ACA increased self-employment among older adults with pre-existing conditions and high demand for insurance. I show that expanded dependent coverage raises self-employment particularly among women and young adults with

diabetes or cancer. These findings suggest that differences in entrepreneurial participation reflect the insurance constraints workers face as well as their preferences for self-employment.

My results also have important policy implications. The insurance value of comprehensive coverage—its ability to reduce large payment risk—matters more than mean subsidies that lower expected health care costs but leave variance unchanged. Policies that mainly top up disposable income while coverage stays tied to a wage job may therefore miss the margin that unlocks entrepreneurship. Portable or catastrophic coverage that compresses the upper tail of health bills is better aligned with the risk-reduction channel I estimate. Mandate-induced self-employment gains concentrate among risk-averse workers, those with high demand for insurance, women, and young adults with diabetes or cancer, so expansions that stabilize health payments can also narrow gaps in who can start a business. Effects among 19–25-year-olds likely understate constraints on older cohorts with higher premiums and morbidity; extending similar protections across the life cycle could further reduce entrepreneurship lock.

2 Institutional Mechanisms

The Affordable Care Act (ACA), enacted in 2010, introduced reforms to expand coverage and improve access to care (Sommers et al., 2013; Garrett and Gangopadhyaya, 2016), and payment and delivery reforms intended to slow health care cost growth (Cutler, 2010). One of the earliest provisions was the dependent coverage mandate, effective in late 2010, which requires private plans to allow dependents to remain on a parent’s policy until age 26 without paying premiums themselves. Before the mandate, most private plans ended dependent coverage at age 19, or at 23 for full-time students (Cantor et al., 2012). The reform therefore extended group-rated coverage to millions of young adults during prime years for labor market entry and business formation; Sommers et al. (2013) estimate that more than three million individuals gained insurance between September 2010 and December 2011. Although some grandfathered plans were initially exempt, the rule applied broadly enough to create a discrete change in insurance access for 19–25-year-olds, a widely used policy shock in health and labor market research (Chua and Sommers, 2014; Barbaresco et al., 2015; Hahn and Yang, 2016; Bailey, 2017; Yörük and Xu, 2019).

For occupational choice, the mandate matters because U.S. health insurance remains largely employment-linked. Wage jobs often bundle subsidized coverage; self-employed workers must purchase individual coverage at higher premiums with greater exposure to out-of-pocket risk (Gruber and Madrian, 2002). When insurance is tied to a parent’s plan rather than the worker’s own job, a young adult can leave wage employment without forfeiting coverage—reducing both expected premiums and the variance of health spending. That institutional change is the identifying variation I use to estimate how expanded insurance access affects self-employment, which subgroups benefit most, and whether responses align with subsidy or risk-reduction mechanisms developed in the conceptual framework.

3 Conceptual Framework

When health insurance is tied to a wage job, coverage can constrain occupational choice and discourage self-employment (Holtz-Eakin et al., 1996; Fairlie et al., 2011). This section develops a mean-variance model of that choice under health expenditure risk. The goal is practical: separate a subsidy channel (expected premiums and out-of-pocket costs) from a risk-reduction channel (dispersion of those payments), and link both objects to the difference-in-differences and two-stage IV designs used later.

Consider an individual choosing between wage employment and self-employment. Wage employment pays wage w and includes employer-sponsored insurance with an implicit subsidy $s^{\text{ESI}} > 0$. The subsidy captures the tax treatment of ESI and the employer’s premium contribution, so group coverage costs less than actuarially equivalent coverage purchased alone. Self-employment yields stochastic profits Π . In the baseline uninsured case, the worker also bears out-of-pocket medical costs m , with no employer risk pooling.

I use mean-variance utility over occupational choice $j \in \{W, S\}$:

$$U_j = E[Y_j] - \frac{\gamma}{2} \text{Var}(Y_j),$$

where Y_j is income net of health expenditures and γ indexes attitudes toward income variance. I do not restrict the sign of γ : it may be positive (risk aversion), zero (risk neutrality), or negative (risk loving).

For wage employment, take net income as deterministic: $Y_W = w - (\pi_I - s^{\text{ESI}})$, where π_I is the actuarial cost of comparable individual coverage. Then $E[Y_W] = w - (\pi_I - s^{\text{ESI}})$ and $\text{Var}(Y_W) = 0$. For uninsured self-employment, uncertainty comes from profits and medical spending: $Y_S = \Pi - m$, so $E[Y_S] = E[\Pi] - \mu_m$ and $\text{Var}(Y_S) = \text{Var}(\Pi) + \text{Var}(m) - 2 \text{Cov}(\Pi, m)$.

The utility gap is

$$U_S - U_W = [E[\Pi] - \mu_m] - [w - (\pi_I - s^{\text{ESI}})] - \frac{\gamma}{2}[\text{Var}(\Pi) + \text{Var}(m) - 2 \text{Cov}(\Pi, m)].$$

Insurance can therefore affect entrepreneurial entry in two ways. On the mean, lacking ESI adds an expected cost $(\pi_I - s^{\text{ESI}}) - \mu_m$ to self-employment—the subsidy disadvantage relative to wage work, plus any gap between premiums and expected out-of-pocket spending. On the variance, uninsured self-employment adds health-expenditure risk $\text{Var}(m)$. How that term enters utility depends on γ . If $\gamma > 0$, variance lowers the value of self-employment beyond the mean cost. If $\gamma = 0$, only expected payments matter. If $\gamma < 0$, extra variance could raise the attractiveness of the uninsured option. I do not take a stand on whether entrepreneurs are risk-averse in general; the empirical object is whether health-payment dispersion predicts occupational choice once expected payments are held fixed.

For eligible young adults aged 19–25, the ACA dependent coverage mandate changes both margins. Parental coverage can cut the expected payment disadvantage of leaving a wage job and compress the dispersion of health bills. With $\gamma = 0$, any response must run through expected payments (or other non-risk channels). With $\gamma > 0$, lower payment variance can relax lock even if mean costs move little. With $\gamma < 0$, mean savings from avoiding individual-market premiums may still dominate any utility loss from reduced variance.

Several implications follow without assuming a single risk type. Relative to slightly older adults who cannot stay on a parent’s plan, eligible young adults should show higher self-employment after the mandate if either channel binds. If the subsidy channel is the main force, responses should look similar across risk profiles. If risk reduction matters, effects should be larger among people with higher health risk or other markers of demand for insurance. The size of the ESI subsidy also matters in principle: where s^{ESI} is

large, neutralizing the wage-job advantage should matter more; where employers offer little coverage, there is less lock to undo. I do not estimate industry-by-industry subsidy schedules here; those contrasts mainly discipline interpretation of the age-based design.

Mandate quasi-experiments also mix selection and treatment. An increase in self-employment can reflect coverage making entry feasible, or a shift in who sorts into self-employment. The empirical sections therefore combine a difference-in-differences comparison of eligible and ineligible ages, behavioral heterogeneity, and a spending-moment decomposition that separates shifts in expected payments from shifts in payment dispersion.

To take the two channels to data, let m denote out-of-pocket medical spending with mean μ_m and variance σ_m^2 . ESI lowers both relative to the uninsured state: $\mu_m^E < \mu_m^U$ and $\sigma_m^{2,E} < \sigma_m^{2,U}$. Parental coverage under the mandate likewise delivers moments $(\mu_m^P, \sigma_m^{2,P})$. The subsidy channel is a first-moment object: $\Delta\mu_m = \mu_m^U - \mu_m^E$ is an income advantage of wage employment that self-employment must overcome. The risk-reduction channel is a second-moment object: insurance compresses $\sigma_m^{2,U}$ toward $\sigma_m^{2,E}$ (or $\sigma_m^{2,P}$). Holding the mean fixed, a response of self-employment to conditional variance is the empirical signature of risk reduction for $\gamma > 0$.

In the data I work with total health payments H : annual out-of-pocket medical spending plus premiums in a person-year (“total pay” in the Methods section). Each person-year contributes one realized H . Realized H^2 is not ex ante variance. For conditioning set $\mathcal{C}$, if $\mu = E[H \mid \mathcal{C}]$, then $E[(H - \mu)^2 \mid \mathcal{C}] = \text{Var}(H \mid \mathcal{C})$. Squared deviations around an estimated conditional mean are therefore a feasible variance proxy.

Let D_i indicate self-employment and let η_i be an i.i.d. taste shock at the occupational threshold:

$$D_i = \mathbb{1}\{f(\mu_i, \sigma_i^2) + \eta_i > 0\}.$$

This motivates the linear probability form used in the IV analysis, $D_i = \beta_0 + \beta_1 \hat{H}_i + \beta_2 \hat{r}_i^2 + \dots + u_i$, where $\hat{H}_i$ is instrumented total pay and $\hat{r}_i^2$ is the instrumented squared residual (the fitted value of $(H_i - \hat{H}_i)^2$). Under correct specification of the conditional-mean and squared-residual equations in the Methods section, these objects proxy expected pay and mandate-induced conditional payment variance. Because H is a cost, $\beta_1 < 0$ is the subsidy-channel prediction: lower expected spending raises self-employment. Holding

$\hat{H}_i$ fixed, the sign of β_v maps to risk preferences: $\beta_v < 0$ is consistent with $\gamma > 0$ (entry falls when the instrumented squared residual is higher); $\beta_v = 0$ with risk neutrality on this margin; and $\beta_v > 0$ with risk-loving tastes ($\gamma < 0$).

The mandate supplies the excluded variation for both instruments. If only expected pay mattered, the instrumented squared residual should be irrelevant once $\hat{H}_i$ is controlled for. If risk reduction matters, self-employment should rise when that instrumented squared residual falls. The later estimates—a negative response to the instrumented squared residual and no precise response to instrumented total pay—are consistent with $\gamma > 0$ under those IV and variance-proxy assumptions. The exercise does not assume that entrepreneurs are more risk-averse than wage workers; it asks whether mandate-induced payment moments move with entry in the direction the model predicts.

The channel distinction also shapes how one reads the policy. If entrepreneurial entry tracked only expected pay, the mandate would look mainly like a transfer financed by other pool members or taxpayers, with efficiency gains depending on entrepreneurial externalities large enough to offset the cost of the transfer. If entry instead tracks lower payment dispersion, parental coverage can relax a market friction—adverse selection, search costs, and thin individual-market options—and raise the value of risky occupational choices even without appealing to spillover arguments. Equity readings differ as well: mean subsidies tilt toward high expected spenders and those who can access generous plans, whereas variance reductions matter most where health bills are hard to forecast and self-insurance is costly. The larger effects among chronically ill young adults and women in the Results section fit that second reading, though they are not themselves a formal test of efficiency.

I do not claim that insurance dominates average-cost subsidies on cost-effectiveness grounds; the design does not compare instruments. The narrower point is that mean transfers and variance reductions are not interchangeable for occupational choice under health risk. Portable or catastrophic coverage that compresses the upper tail of health payments can move entry through a margin that average-cost subsidies miss. Measuring aggregate efficiency gains from that margin is outside the present design.

4 Data

I draw on the Medical Expenditure Panel Survey (MEPS), linking the Household Component with the Person Round Plan files to match employment status to plan-level health insurance characteristics. The final analysis sample contains 47,125 person-year observations for 2006–2014. MEPS is a nationally representative survey of the U.S. civilian noninstitutionalized population conducted by the Agency for Healthcare Research and Quality (AHRQ), with rich data on demographics, socioeconomic status, health, insurance, access to care, and medical expenditures. IPUMS MEPS provides the complex-survey design variables needed for design-consistent inference: person-level weights (`PERWEIGHT`), primary sampling units (`PSUANN`), and strata (`STRATANN`).

MEPS uses an overlapping panel design: each year a new cohort is interviewed over five rounds spanning two calendar years, which yields the pooled person-year observations I analyze for 2006–2014. That window covers the years immediately before and after the 2010 dependent coverage mandate, which allowed young adults to remain on a parent’s private health insurance plan until age 26. I define the treatment group as individuals aged 19 to 25 and the control group as those aged 27 to 34; 26-year-olds are partially treated. The sample ends in 2014, before platform gig work (e.g., Uber) became widespread nationally.

The final dataset contains 47,125 person-year observations after applying the age restriction and dropping observations with missing values in key variables such as employment status, health insurance coverage, and demographic characteristics. MEPS reports SF-12 Physical Component Summary (PCS) and Mental Component Summary (MCS) scores; higher scores indicate better health. Heterogeneity tables code below-median PCS or MCS as higher health risk. The main outcome variable is an indicator for self-employment, defined in MEPS as being self-employed in either an incorporated or unincorporated entity, including sole proprietorship and partnership. This definition aligns with broader labor economics literature and reflects self-employment status as reported by individuals, regardless of business registration or legal structure.

A limitation of this measure is that MEPS does not separate traditional business founders from gig-economy contractors. Recent work using U.S. tax returns shows that many gig workers later transition into entrepreneurship, often through small sole proprietorships in industries related to their gig experience (Denes et al., 2025). In MEPS I observe le-

gal form (incorporated, unincorporated, partnership) and self-reported self-employment, but not business income, platform affiliation, or whether earnings come from contract platforms (e.g., Uber), delivery, or freelance work. Some young adults classified as self-employed may therefore be entering gig-style activity rather than employer firms with employees. The estimates should be read as effects on broad self-employment and non-employer business activity, not on high-growth entrepreneurship alone.

Table 1 summarizes baseline characteristics. Young adults in the treatment group have a self-employment rate about 5 pp lower than the control group (t -statistic 21), so the treated cohort starts from a lower entrepreneurial baseline. Large t -statistics on age, gender, education, marital status, college enrollment, and race confirm demographic differences between groups; health measures also differ. Because the sample spans nine years, macroeconomic shocks such as inflation, minimum wages, and the business cycle could move self-employment independently of the mandate. Year fixed effects and region controls in the empirical strategy address these confounders.

Table 1: Summary Statistics by Treatment

Variable	Full Sample		Treated		Not Treated		Diff. Means	
	Mean	St. Dev.	Mean	St. Dev.	Mean	St. Dev.	Diff.	t
Self-employed	0.062	0.240	0.035	0.185	0.083	0.276	-0.048	-21.200
Year	2010.143	2.530	2010.146	2.662	2010.143	2.536	0.003	0.099
Age	26.680	4.590	22.054	1.997	30.534	2.277	-8.480	-405.472
Sex	0.492	0.500	0.476	0.499	0.506	0.500	-0.030	-6.038
Education	1.976	1.620	1.509	1.297	2.323	1.750	-0.814	-54.817
In College	0.148	0.355	0.350	0.477	0.000	0.000	0.350	97.883
Marital Status	3.484	1.893	4.444	1.365	2.686	1.899	1.757	110.467
Race	0.523	1.006	0.529	1.009	0.515	1.004	0.014	1.442
Hispanic	0.325	0.468	0.334	0.472	0.321	0.467	0.013	2.729
Family Size	3.373	1.776	3.440	1.828	3.370	1.730	0.069	3.940
Diabetes	0.012	0.108	0.005	0.073	0.018	0.131	-0.012	-12.159
PCS	54.484	7.695	54.849	7.278	54.151	8.038	0.698	9.298
MCS	52.040	10.199	52.312	10.286	51.802	10.158	0.510	5.054
Regions	2.834	0.984	2.828	0.984	2.841	0.986	-0.014	-1.413
N	47,125		17,778		24,333		-	

Note: This table presents the summary statistics for the treated and non-treated groups. Since 26-year-olds are partially treated, they are excluded from both the treated and control groups.

5 Methods

I estimate the effect of the dependent coverage mandate on self-employment using a difference-in-differences (DiD) design. The policy shock expands parental coverage for young adults aged 19–25 relative to slightly older adults who remain ineligible. The estimand is the change in self-employment attributable to mandate eligibility. DiD differences out time-invariant gaps between the young and older cohorts and common year-to-year shocks, including macro conditions after the Great Recession. Every specification includes year fixed effects and age fixed effects. The benchmark model reported in Table 2 further absorbs fixed effects for demographic observables: region, gender, race, Hispanic ethnicity, education, marital status, family size, and college enrollment. The treatment coefficient therefore reflects differential post-2010 changes among observationally similar individuals rather than compositional differences across groups.

The treated group consists of ages 19–25; the control group consists of ages 27–34. I code 26-year-olds as partially treated because eligibility during rollout depended on plan enrollment and timing. The baseline specification is a linear probability model:

$$\text{Self-employment}_{it} = \beta_1 \cdot \text{Treatment}_{it} + \beta_2' \mathbf{X}_{it} + \tau_t + \epsilon_{it} \quad (1)$$

where Treatment_{it} equals one for mandate-eligible ages in post-2010 years (equivalently, $\text{POST}_t \times \text{TREATED}_i$). In the implementation, age and year enter as fixed effects in all models (τ_t denotes year effects). In Table 2, the demographic observables listed above also enter as fixed effects through $\mathbf{X}_{it}$. Other robustness checks (event study, placebo ages, nested age-window trims, a control for spousal employer coverage, survey-weighted replication, logit and probit models) keep this same fixed-effect structure; only the specification-curve analysis varies which demographic fixed effects are included, always holding age and year fixed.

The benchmark models in Table 2 are unweighted. MEPS is a stratified, multistage probability sample; a design-consistent alternative applies person weights (`PERWEIGHT`) with primary sampling units (`PSUANN`) and strata (`STRATANN`) on the full file and restricts inference to ages 19–34 as a subpopulation (Korn and Graubard, 1999; Lumley, 2010). For a causal DiD estimand, survey weights are not required by default: the relevant question is why one would weight (Solon et al., 2015). I use an unweighted baseline because MEPS oversampling is driven mainly by race and ethnicity (and, in some panels, pre-

dicted poverty)—domains absorbed by the demographic fixed effects—while current self-employment does not determine sample inclusion; when stratification is based on exogenous variables, unweighted estimation with those controls can be consistent (Wooldridge, 1999). The DiD parameter compares cohorts before and after 2010, so I cluster standard errors two-way on age and year, which matches treatment assignment. Following Solon et al. (2015), I also report design-weighted estimates as a diagnostic in Online Appendix Table A3; Results discuss that comparison.

The pooled DiD estimate assumes that, absent the mandate, self-employment among 19–25-year-olds would have followed the same trend as among 27–34-year-olds. I test that assumption and trace the timing of mandate-linked entrepreneurial entry using an event study. For each year from 2006 to 2014 (omitting 2010 as the reference year, when the dependent coverage rule took effect), the specification compares the self-employment rate of young adults to that of the older control group, holding fixed the same age, year, and demographic fixed effects as in Table 2. If entrepreneurship lock binds through this policy shock, coefficients should be near zero before 2010 and rise after coverage expands for young adults but not for the control cohort. The post-2010 path also shows whether the effect appears immediately in 2010 or builds over time, consistent with gradual take-up of dependent coverage and adjustment from wage employment to self-employment.

$$\text{Self-employment}_{it} = \sum_{k=2006, k \neq 2010}^{2014} \beta_k (\text{Year}_k \times \text{Young adult}_i) + \delta' \mathbf{X}_{it} + \tau_t + \epsilon_{it} \quad (2)$$

The behavioral heterogeneity results suggest that mandate-induced changes in health spending matter through insurance value, not only through the level of total pay. To test that distinction on the full sample, I use a two-stage fitted-value IV procedure with principal component analysis (PCA). The key spending measure is total pay: annual out-of-pocket medical spending plus premiums in a person-year,

$$\text{TotalPay}_{it} = \text{EXPSELFAY}_{it} + \overline{\text{OOPPREM}}_{it},$$

where EXPSELFAY_{it} is annual out-of-pocket spending on medical care (self/family payments, IPUMS annual person file) and $\overline{\text{OOPPREM}}_{it}$ is the within-year average of annual out-of-pocket premiums. Premiums come from the MEPS Person Round Plan variable OOPX12X (annual out-of-pocket premium amount for the round, in dollars), averaged

within person-year after policyholder/dependent recodes (2006–2014), then collapsed to the person-year. Total pay is measured in thousands of dollars and mean-centered. Denote $H_{it} = \text{TotalPay}_{it}$.

A single observed annual payment is one draw from a person’s health-payment distribution, not that person’s ex ante spending risk. In particular, realized H_{it}^2 is not the variance of health spending. The variance object in the mean-variance model is the dispersion of possible payments before occupational choice. Let $\mathcal{C}_i$ denote the conditioning information in the payment equations (mandate variation, age, year, and demographic controls). If $\mu_i = E[H_i | \mathcal{C}_i]$, then

$$E[(H_i - \mu_i)^2 | \mathcal{C}_i] = \text{Var}(H_i | \mathcal{C}_i),$$

so a squared deviation around the true conditional mean is a noisy observation whose conditional expectation equals conditional payment variance. In the feasible procedure, I first instrument realized total pay and use its fitted value as an estimate of the conditional mean. I then form squared residuals around that fitted value and instrument those squared residuals with mandate-induced variation. Under correct specification of the conditional-mean equation, the fitted squared-residual component can be interpreted as mandate-induced variation in conditional payment variance. The second stage asks whether self-employment responds to fitted total pay (subsidy channel) or to this fitted squared-residual component (risk-reduction channel).

I first apply PCA to eight baseline categorical covariates: region, marital status, race, Hispanic ethnicity, education, student status, family size, and gender. Let $\mathbf{Z}_i$ denote these covariates for individual i . PCA transforms $\mathbf{Z}_i$ into orthogonal components

$$\mathbf{PC}_i = \mathbf{W}'\mathbf{Z}_i,$$

where $\mathbf{PC}_i = (PC_{i1}, PC_{i2}, PC_{i3}, PC_{i4})'$ summarizes baseline heterogeneity in $\mathbf{Z}_i$. These components allow mandate instruments to absorb type-specific treatment effects without a separate interaction for every category.

To construct the instrument for realized total pay, I estimate unrestricted and restricted auxiliary regressions of total pay on the mandate:

$$\text{TotalPay}_i = \gamma_0 + \gamma_1 T_i + \sum_{k=1}^4 \gamma_{1+k} PC_{ik} + \sum_{k=1}^4 \gamma_{5+k} (T_i \times PC_{ik}) + \alpha_{\text{AGE}(i)} + \delta_{\text{YEAR}(i)} + \epsilon_i, \quad (3)$$

$$\text{TotalPay}_i = \eta_0 + \sum_{k=1}^4 \eta_{1+k} PC_{ik} + \alpha_{\text{AGE}(i)} + \delta_{\text{YEAR}(i)} + \xi_i. \quad (4)$$

The total-pay instrument is the difference in fitted values,

$$\text{TotalPay}_i^{\text{IV}} = \text{TotalPay}_i^{\text{unrestricted}} - \text{TotalPay}_i^{\text{restricted}}.$$

The restricted fit omits the mandate terms, while the unrestricted fit permits mandate-induced shifts that differ across demographic profiles. Their difference isolates the component of predicted total pay associated with the mandate and its interactions, conditional on the included age, year, and demographic controls.

The first stage for total pay is

$$\text{TotalPay}_i = \pi_0 + \pi_1 \text{TotalPay}_i^{\text{IV}} + \alpha_{\text{AGE}(i)} + \delta_{\text{YEAR}(i)} + \sum_j \phi_j \lambda_j(\mathbf{Z}_i) + \varepsilon_i, \quad (5)$$

where $\lambda_j(\mathbf{Z}_i)$ denotes category fixed effects. Let $\hat{H}_i$ denote the fitted value from (5). I interpret $\hat{H}_i$ as an estimate of the conditional mean μ_i under correct specification of this first stage.

I then form the squared residual $r_i^2 = (H_i - \hat{H}_i)^2$. Because insurance can compress the spread of possible bills—especially catastrophic ones—the mandate may shift r_i^2 differentially across types even when fitted total pay moves little. Take two cases. A young adult with predictable maintenance costs (e.g., chronic asthma therapy) has small payment surprises; parental coverage may primarily lower the typical bill. A young adult facing a concentrated high-cost risk (e.g., an uninsured accident or hospital stay) has large surprises; parental coverage may primarily reduce dispersion. Interacting the mandate with principal components permits the squared-residual auxiliaries to capture such heterogeneous shifts.

I construct a squared-residual instrument exactly as for total pay, replacing H_i with r_i^2 in unrestricted and restricted auxiliaries, and define

$$r_i^{2,IV} = \hat{r}_i^{2,\text{unrestricted}} - \hat{r}_i^{2,\text{restricted}}.$$

The first stage for the squared residual is

$$r_i^2 = \rho_0 + \rho_1 r_i^{2,IV} + \alpha_{\text{AGE}(i)} + \delta_{\text{YEAR}(i)} + \sum_j \psi_j \lambda_j(\mathbf{Z}_i) + \eta_i. \quad (6)$$

Let $\hat{r}_i^2$ denote the fitted value from (6). Although the observed outcome r_i^2 is nonnegative, a linear fitted value need not be nonnegative. Under correct specification of the conditional-mean and squared-residual equations, $\hat{r}_i^2$ is interpreted as the mandate-induced component of conditional payment variance; without those conditions, it is more narrowly a fitted squared-residual measure.

The second stage links self-employment to fitted total pay and the fitted squared-residual measure:

$$\text{Self-employment}_i = \beta_0 + \beta_1 \hat{H}_i + \beta_v \hat{r}_i^2 + \alpha_{\text{AGE}(i)} + \delta_{\text{YEAR}(i)} + \sum_j \kappa_j \lambda_j(\mathbf{Z}_i) + u_i.$$

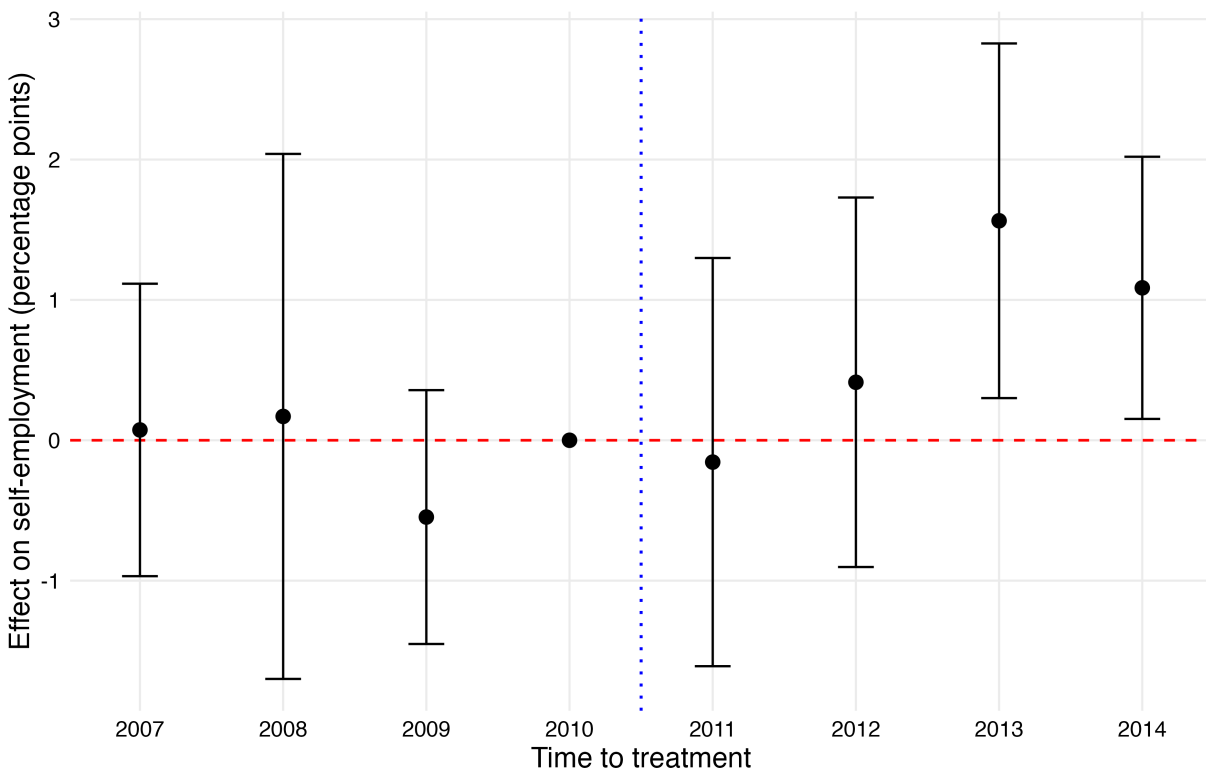
Total pay is a cost: lower fitted total pay should raise self-employment if the subsidy channel dominates ($\beta_1 < 0$). Under the variance interpretation above, a negative β_v indicates that self-employment rises when the mandate-induced component of conditional payment variance falls, holding fitted total pay fixed.

Because $\hat{H}_i$ and $\hat{r}_i^2$ are generated regressors, conventional second-stage standard errors do not account for uncertainty from the preceding estimation steps. I therefore report second-stage standard errors from a cluster bootstrap over age–year cells that re-estimates the entire pipeline—PCA, auxiliary mandate regressions, instruments, both first stages, and the second stage—in each replication. Second-stage coefficients are reported in percentage points in Table 4. The table also reports equation-specific first-stage F -statistics; these summarize the relevance of each constructed instrument in its corresponding first stage but are not joint weak-identification tests for the two-regressor system. Online Appendix Section B evaluates the residual-variance interpretation in a Monte Carlo with known latent σ_i^2 .

6 Results

I begin with an event-study specification to assess parallel pre-trends and dynamic treatment effects. Figure 1 plots year-by-year estimates for young adults relative to the control group. Pre-2010 coefficients cluster around zero with overlapping confidence intervals, supporting the parallel-trends assumption. After 2010, estimates turn positive and precision improves; point estimates rise gradually through 2013–2014, consistent with gradual take-up of dependent coverage and time to switch from wage employment to self-employment after the rule took effect.

Figure 1: Event Study Plot



Next, I estimate the benchmark difference-in-differences model as a linear probability model with the full set of fixed effects used in the specification curve (age, year, region, gender, marital status, family size, race, Hispanic ethnicity, education, and college enrollment; 256 combinations in total). Table 2 reports the preferred specification. The treatment coefficient is 0.89 pp ($p < 0.01$), a 26% increase relative to the 2009 treated-group

mean self-employment rate. The effect is economically meaningful: entrepreneurship lock is present, and expanded dependent coverage materially raises young adults' self-employment.

Table 2: Difference-in-Differences Estimation

	<i>Dependent Variable:</i>
	Self-employment rate
Treatment	0.892*** (0.336)
Implied percentage change	25.935
Fixed Effects	Yes
Observations	47,125

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Fixed effects include age, year, region, gender, marital status, family size, race, Hispanic ethnicity, educational attainment, and college student status. Robust standard errors clustered two-way by age and year are given in parentheses.

I subject the headline estimate to several additional robustness checks; tables and figures are in the Online Appendix. Table A3 re-estimates the benchmark model using MEPS person weights and design-based standard errors on the full survey file with a subpopulation restriction to ages 19–34, following the diagnostic comparison recommended by Solon et al. (2015). The treatment coefficient is 0.86 pp, nearly the same as the unweighted 0.89 pp, but the standard error rises from 0.34 to 0.60 and the estimate is no longer significant at conventional levels ($p = 0.15$). The mandate effect is stable in magnitude; accounting for the complex survey design mainly adds variance rather than shifting the point estimate.

Table A4 re-estimates the benchmark specification as logit and probit models with the same fixed effects. Average marginal effects for treatment are 0.65 pp in both models ($p < 0.10$), positive and in the same direction as the linear probability estimate, so the headline result is not an artifact of the LPM functional form.

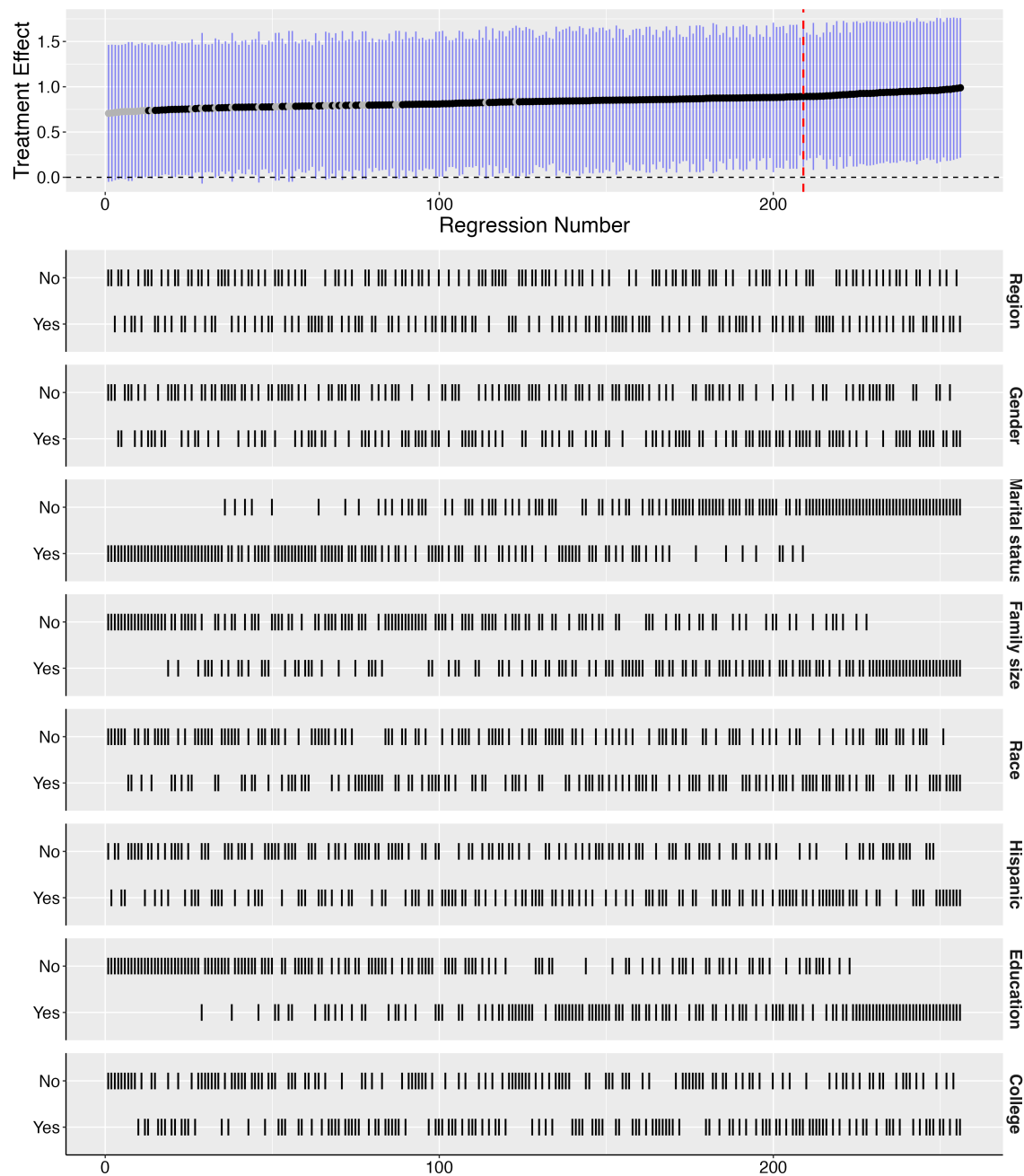
For placebo ages, I redefine the treated group as ages 27–34 and the control as ages 35–42 (Table A1). Neither cohort was directly affected by dependent coverage. The placebo coefficient is effectively zero with large standard errors, so I detect no discernible change in self-employment when comparing placebo cohorts.

A concern for any age-based DiD is that the effect reflects broad differences between young and slightly older workers rather than mandate eligibility at the boundary. I therefore re-estimate the benchmark model while progressively narrowing the treated and control age ranges so the two groups are closer in age (Figure A1). Bandwidth 8 reproduces the main specification; at bandwidth 1, only 25-year-olds are treated and only 27-year-olds serve as controls (age-window definitions are in the figure notes). Estimates remain positive and similar in magnitude across windows, indicating that the main result is not driven by comparing very young adults to much older controls.

The mandate expanded dependent coverage on parental plans, but some young adults may instead obtain insurance through a spouse’s employer. Prior work uses spousal coverage as an alternative route out of job lock, so part of the main estimate could reflect marriage-linked insurance rather than the dependent coverage rule. The mandate itself applies regardless of marital status, but spousal coverage could still correlate with self-employment and with treatment intensity. Table A2 adds an indicator for coverage through a spouse’s or deceased spouse’s previous employer (MEPS establishment type 10). The treatment coefficient remains 0.89 pp ($p < 0.01$), close to the benchmark estimate, while spousal coverage enters negatively and significantly. The mandate effect is therefore not explained by substitution into spousal employer plans.

As a further check that the headline DiD estimate is not an artifact of one fixed-effect bundle, Figure 2 displays a specification curve over the same 256 demographic fixed-effect combinations. The top panel plots treatment estimates sorted by size, with blue bars for confidence intervals; black points mark specifications whose intervals exclude zero, and grey points those that do not. The bottom panels record, for each regression, whether each demographic fixed effect is included. Estimated effects are positive across all models. The red dashed line marks the preferred specification with the full set of demographic fixed effects (Table 2); that benchmark lies well within the distribution of positive estimates, indicating that the headline effect is not driven by a particular fixed-effect choice.

Figure 2: Specification Curve



Thus far, the mandate increased self-employment among eligible young adults. The conceptual framework implies that if risk reduction rather than mean subsidies drives the response, treatment effects should concentrate among risk-averse individuals and those with high demand for insurance (Kihlstrom and Laffont, 1979; Hvide and Panos, 2014; Skriabikova et al., 2014; Hochmuth et al., 2025). Table 3 reports sanity checks on that prediction. They are not separate identifying assumptions, but they ask whether mandate effects line up with behaviorally predicted patterns. In Column (1), risk-averse individuals are 1.5 pp less likely to be self-employed pre-mandate, yet the treatment $\times$ risk-aversion interaction is 1.6 pp ($p < 0.05$), so the average mandate effect among risk-averse young adults is positive while risk-tolerant types show no response. In Column (2), high insurance demand is associated with 2.5 pp lower baseline self-employment, but a 1.5 pp positive interaction ($p < 0.05$) implies that mandate gains are concentrated among those who value coverage most. The signs are consistent with the risk-reduction channel, though these interactions do not by themselves identify mechanisms.

Table 3: Heterogeneity Analysis: Behavioral Factors

	Risk Aversion (1)	Health Insurance Demand (2)
Treatment	0.062 (0.342)	0.081 (0.551)
Behavioral Factor	-1.478*** (0.210)	-2.498*** (0.408)
Treatment $\times$ Factor	1.590*** (0.162)	1.463*** (0.478)
Fixed Effects	Yes	Yes
Observations	45,937	45,937

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. The dependent variable is the self-employment rate. Risk aversion is coded as 1 for individuals with high risk aversion, and 0 for those with low risk aversion. Health insurance demand is coded as 1 for individuals with high demand, and 0 for those with low demand. Fixed effects include age, year, region, gender, marital status, family size, race, Hispanic ethnicity, educational attainment, and college student status. Coefficients are scaled by 100 and interpreted as percentage points. Robust standard errors clustered two-way by age and year are given in parentheses.

I formalize this distinction with a two-stage IV procedure for realized total pay and the squared residual around its instrumented value (Table 4). First-stage F -statistics are 42.34 and 10.49 (Columns 1–2). In the second stage (Column 3), the coefficient on instrumented total pay is small and imprecise (-0.26 pp), while the coefficient on the instrumented squared residual is -0.98 pp (bootstrap SE = 0.60, $p = 0.099$). Because total pay is measured in thousands of dollars, the latter coefficient is the change in self-employment percentage points associated with a one-unit change in variance measured in $(\$1,000)^2$. Thus, self-employment rises as the instrumented squared residual falls, holding instrumented total pay fixed.

If the total-pay first stage recovers the relevant conditional mean and the squared-residual equation is correctly specified, the instrumented squared residual measures conditional payment variance; under that interpretation, the pattern is consistent with the risk-reduction channel rather than with a subsidy channel operating through expected pay alone. The imprecise coefficient on instrumented total pay does not imply that the mandate left average costs unchanged; it indicates that entry does not track that mean component once the instrumented squared residual is included. Online Appendix Section B evaluates that variance interpretation in a Monte Carlo that compares an oracle variance regressor, OLS on realized H^2 , and the squared-residual IV procedure under known latent σ_i^2 .

These estimates sharpen the policy takeaway from the DiD. Because entrepreneurial entry tracks the risk-reduction channel rather than the subsidy channel, reforms judged only by how much they cut expected out-of-pocket health care costs may understate their effect on business formation if they leave workers exposed to rare but large medical bills. Mean subsidies that keep the job–plan link but trim average costs can look generous on paper while doing little to relax entrepreneurship lock.

I also explored interacting instrumented total pay and the instrumented squared residual with physical and mental health, specific health conditions (e.g., diabetes and cancer), and behavioral factors (risk aversion and demand for health insurance); those exercises are underpowered, and I leave a systematic IV heterogeneity analysis to future work.

Finally, I examine distributional consequences. Rietveld et al. (2015b) document positive selection of healthier individuals into self-employment; expanding dependent coverage should matter most for those facing high health risk. I measure heterogeneity using SF-12 Physical Component Summary (PCS) and Mental Component Summary (MCS)

Table 4: Two-Stage IV Estimates: Total Pay and the Squared-Residual Component

	<i>Dependent Variable</i>		
	Total Pay	Squared Residual	Self-Employment Rate
	(1) First Stage	(2) First Stage	(3) Second Stage
Instrument	0.878*** (0.135)	1.056** (0.326)	
Instrumented total pay			−0.26 (6.45)
Instrumented squared residual			−0.98* (0.60)
Observations	47,125	47,125	47,125
Fixed Effects	Yes	Yes	Yes
First-Stage <i>F</i> -Statistic	42.34	10.49	

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Columns (1)–(2) report first-stage regressions of realized total pay and of the squared residual around instrumented total pay, $(H - \hat{H})^2$, on their respective instruments. Total pay is annual out-of-pocket medical spending plus premiums (\$1,000s; defined in the Methods section). Column (3) reports the second-stage regression of self-employment on instrumented total pay and the instrumented squared residual. Under correct specification of the total-pay and squared-residual equations, these objects proxy expected pay and the mandate-induced component of conditional payment variance. Fixed effects include age, year, region, gender, marital status, family size, race, Hispanic ethnicity, educational attainment, and college student status. First-stage standard errors are clustered by age–year pairs. Reported *F*-statistics are equation-specific and are not joint weak-identification tests for the two-regressor system. Second-stage standard errors in Column (3) are from a 200-replication cluster bootstrap over age–year cells that re-estimates the entire pipeline (PCA, auxiliary mandate regressions, instruments, first stages, and second stage), accounting for generated regressors. Second-stage coefficients are scaled by 100 (percentage points).

scores (higher scores indicate better health) and indicators for chronic conditions (Table 5). Panel A (Columns 1–2): healthier individuals (higher PCS) are 0.51 pp more likely to be self-employed, in line with Rietveld et al. (2015a) and Yoon and Bernell (2013). Treatment interactions with PCS and MCS are imprecise, so the average mandate effect does not differ detectably by general physical or mental health scores. Panel B focuses on chronic conditions. Young adults with diabetes are 3.8 pp less likely to be self-employed at baseline (Column 1). The treatment $\times$ diabetes interaction is 5.0 pp ($p < 0.05$), implying a large mandate-induced gain for diabetics. For cancer (Column 2), the interaction is 3.4 pp ($p < 0.01$). I find no precise differential effects for hypertension or cardiovascular disease (Columns 3–4), likely reflecting limited power. The mandate therefore relaxes entrepreneurship lock most clearly for young adults with high-cost chronic conditions.

Table 5: Heterogeneity Analysis: Health Conditions

<i>Panel A: General Health</i>				
	Physical Health	Mental Health		
	(1)	(2)		
Treatment	0.844*** (0.257)	1.071*** (0.325)		
PCS / MCS	-0.505*** (0.189)	0.027 (0.092)		
Treatment × PCS / MCS	0.14 (0.248)	-0.418 (0.331)		
Fixed Effects	Yes	Yes		
Observations	47,125	47,125		
<i>Panel B: Specific Health Conditions</i>				
	Diabetes	Cancer	Hypertension	Cardiovascular Disease
	(1)	(2)	(3)	(4)
Treatment	0.866*** (0.328)	0.752*** (0.28)	0.699** (0.319)	0.708** (0.359)
Health Condition	-3.794*** (0.662)	-0.815 (1.312)	-1.236*** (0.451)	0.242 (0.757)
Treatment × Health Condition	4.982** (2.388)	3.379*** (1.256)	0.523 (0.934)	0.49 (0.702)
Fixed Effects	Yes	Yes	Yes	Yes
Observations	47,125	37,943	42,368	42,368

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. The dependent variable is the self-employment rate. Diabetes, cancer, hypertension, and cardiovascular disease are coded as 1 for individuals with the respective condition and 0 otherwise. PCS and MCS are coded 0 if the score is above the median and 1 otherwise. Fixed effects include age, year, region, gender, marital status, family size, race, Hispanic ethnicity, educational attainment, and college student status. Coefficients are scaled by 100 (percentage points). Standard errors clustered two-way by age and year are in parentheses.

Motivated by Blume-Kohout (2023), I next interact treatment with gender (Table 6). Before 2010, women are 3.2 pp less likely than men to be self-employed (Cortes and Pan, 2018; Shurchkov and Eckel, 2018), partly reflecting higher risk aversion (Cohen and Einav, 2007). The treatment×female coefficient is 2.3 pp ($p < 0.01$): women experience a larger mandate-induced increase in self-employment than men, consistent with entrepreneurship lock binding more where insurance value is high. Policy wise, dependent coverage expands insurance access and narrows gender gaps in entrepreneurial entry among young adults.

Table 6: Gender Equity Analysis

	<i>Self-employment rate</i>
Treatment	−0.217 (0.143)
Female	−3.236*** (0.329)
Treatment × Female	2.332*** (0.315)
Fixed Effects	Yes
Observations	47,125

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Fixed effects include age, year, region, gender, marital status, family size, race, Hispanic ethnicity, educational attainment, and college student status. Standard errors clustered two-way by age and year are in parentheses.

7 Conclusions

This paper asks whether entrepreneurship lock binds for young adults when health insurance is tied to employment. Using the ACA dependent coverage mandate and MEPS data in a difference-in-differences design, I find that eligibility raised self-employment by 0.89 pp—about 26% relative to the pre-mandate treated mean. Event-study estimates show flat pre-trends and positive post-2010 effects; placebo cohorts show no response. The magnitude is robust across 256 fixed-effect specifications, nested age-window trims, and a control for spousal employer coverage.

These sanity checks and the IV estimates are consistent with insurance value rather than income subsidies. Treatment interactions concentrate among risk-averse individuals and those with high demand for coverage. Self-employment shows no discernible response to instrumented total pay but rises as the instrumented squared residual falls (−0.98 pp, bootstrap SE = 0.60, $p = 0.099$). Under the conditional-mean and variance-proxy assumptions in the Methods section, this pattern is consistent with risk reduction in the mean-variance framework. Heterogeneity analyses show especially large gains for young adults with diabetes or cancer and an additional 2.3 pp increase for women relative to men.

These findings advance three literatures: health-related job lock and entrepreneurial supply; empirical efforts to distinguish expected-cost and risk-reduction channels; and distributional consequences of coverage expansions for chronically ill workers and women. They also suggest that comprehensive, portable coverage can promote entrepreneurship by reducing financial exposure beyond its effect on expected payments.

Several limitations remain. The design identifies effects for ages 19–25; older workers with higher expected costs may face stronger lock, so extrapolation requires caution. MEPS self-employment is a broad, self-reported measure that cannot distinguish incorporated startups, low-growth sole proprietorships, and gig-economy contractors (Denes et al., 2025). I do not observe business income or platform earnings, so mandate-induced entries may include contract work (e.g., Uber) as well as traditional entrepreneurship. Main inference uses unweighted models with two-way clustering on age and year; a replication that uses survey weights is similar in magnitude (0.86 versus 0.89 pp) but less precise once the complex design is fully accounted for. I do not estimate IV heterogeneity by subgroup because of power constraints.

Future work should situate this estimate in the broader landscape of U.S. entrepreneurship. I identify effects for a young, relatively healthy cohort using a partial-equilibrium design and a broad self-employment measure. If insurance frictions bind more strongly for older workers, for transitions into employer firms, or for high-growth ventures not captured in household surveys, the economy-wide cost of entrepreneurship lock could exceed what I find here. Linking administrative business-formation and insurance records would help separate extensive-margin entry from changes in firm quality. A complementary agenda is to ask whether policies that reduce conditional variance in background risk, not only mean transfers, shift occupational choice in other settings where background risk and liquidity constraints interact, such as retirement, housing, or consumer credit. Structural models that keep mean and variance effects distinct could quantify how much talent remains locked when insurance stays tied to a single job.

The dependent coverage mandate is one illustration: when coverage can move with the worker, more young adults enter self-employment, and the response is largest where risk reduction matters most. As nontraditional work grows, understanding how institutions that stabilize consumption or earnings risk shape who can start a business will remain central for labor supply and for measuring the social cost of lock beyond this age group.

References

- Acs, Z. J. and Audretsch, D. B. (1988). Innovation in large and small firms: An empirical analysis. American Economic Review, 78(4):678–690.
- Aghion, P. and Howitt, P. (1992). A model of growth through creative destruction. Econometrica, 60(2):323–351.
- Bailey, J. (2017). Health insurance and the supply of entrepreneurs: New evidence from the affordable care act. Small Business Economics, 49(3):627–646.
- Bailey, J. and Dave, D. (2019). The effect of the affordable care act on entrepreneurship among older adults. Eastern Economic Journal, 45:141–159.
- Barbaresco, S., Courtemanche, C. J., and Qi, Y. (2015). Impacts of the affordable care act dependent coverage provision on health-related outcomes of young adults. Journal of health economics, 40:54–68.
- Baumol, W. J. (1990). Entrepreneurship: Productive, unproductive, and destructive. Journal of Political Economy, 98(5):893–921.
- Blume-Kohout, M. E. (2023). The affordable care act and women’s self-employment in the united states. Feminist Economics, 29(1):174–204.
- Blume-Kohout, M. E. (2024). Entrepreneurship lock and the demand for health insurance: Evidence from the us affordable care act. ILR Review, 77(2):199–226.
- Cantor, J. C., Monheit, A. C., DeLia, D., and Lloyd, K. (2012). Early impact of the affordable care act on health insurance coverage of young adults. Health Services Research, 47(5):1773–1790.
- Chua, K.-P. and Sommers, B. D. (2014). Changes in health and medical spending among young adults under health reform. Jama, 311(23):2437–2439.
- Cohen, A. and Einav, L. (2007). Estimating risk preferences from deductible choice. American economic review, 97(3):745–788.
- Cortes, P. and Pan, J. (2018). Occupation and gender. The Oxford handbook of women and the economy, pages 425–452.
- Cutler, D. M. (2010). How health care reform must bend the cost curve. Health Affairs, 29(6):1131–1135.
- DeCicca, P. (2010). Health insurance availability and entrepreneurship.
- Denes, M., Lagaras, S., and Tsoutsoura, M. (2025). Entrepreneurship and the Gig economy: Evidence from U.S. tax returns. Journal of Financial Economics, 173:104156.
- Fairlie, R. W., Kapur, K., and Gates, S. (2011). Is employer-based health insurance a barrier to entrepreneurship? Journal of Health Economics, 30(1):146–162.
- Garrett, A. B. and Gangopadhyaya, A. (2016). Who gained health insurance coverage under the aca, and where do they live? Urban Institute, ACA implementation—monitoring and tracking.

- Garthwaite, C., Gross, T., and Notowidigdo, M. J. (2014). Public health insurance, labor supply, and employment lock. The Quarterly Journal of Economics, 129(2):653–696.
- Glaeser, E. L., Kerr, S. P., and Kerr, W. R. (2015). Entrepreneurship and urban growth: An empirical assessment with historical mines. Review of Economics and Statistics, 97(2):498–520.
- Gruber, J. and Madrian, B. C. (1994). Health insurance and job mobility: The effects of public policy on job-lock. ILR Review, 48(1):86–102.
- Gruber, J. and Madrian, B. C. (2002). Health insurance, labor supply, and job mobility: A critical review of the literature.
- Hahn, Y. and Yang, H.-S. (2016). Do work decisions among young adults respond to extended dependent coverage? ILR Review, 69(3):737–771.
- Haltiwanger, J. C., Jarmin, R. S., and Miranda, J. (2013). Who creates jobs? small vs. large vs. young. Review of Economics and Statistics, 95(2):347–361.
- Hamersma, S. and Kim, M. (2009). The effect of parental medicaid expansions on job mobility. Journal of health economics, 28(4):761–770.
- Heim, B. T. and Yang, L. K. (2017). The impact of the affordable care act on self-employment. Health economics, 26(12):e256–e273.
- Hochmuth, B., Merz, M., and Prettenhaler, F. (2025). Heterogeneous risk preferences, entrepreneurship, and wealth. CEPR Discussion Paper DP20286, Centre for Economic Policy Research.
- Holtz-Eakin, D., Penrod, J. R., and Rosen, H. S. (1996). Health insurance and the supply of entrepreneurs. Journal of Public Economics, 62(1-2):209–235.
- Hvide, H. and Panos, G. (2014). Risk tolerance and entrepreneurship. Journal of Financial Economics, 111(1):200–223.
- IPUMS MEPS (2021). Integrated public use microdata series, medical expenditure panel survey (ipums meps), version 2.1 [data set].
- Kihlstrom, R. E. and Laffont, J.-J. (1979). A general equilibrium entrepreneurial theory of firm formation based on risk aversion. Journal of Political Economy, 87(4):719–748.
- Korn, E. L. and Graubard, B. I. (1999). Analysis of Health Surveys. Wiley, New York.
- Lumley, T. (2010). Complex Surveys: A Guide to Analysis Using R. Wiley, Hoboken, NJ.
- Madrian, B. C. (1994). Employment-based health insurance and job mobility: Is there evidence of job-lock? The Quarterly Journal of Economics, 109(1):27–54.
- Murphy, K. M., Shleifer, A., and Vishny, R. W. (1991). The allocation of talent: Implications for growth. Quarterly Journal of Economics, 106(2):503–530.
- Niu, X. (2014). Health insurance and self-employment: Evidence from massachusetts. ILR Review, 67(4):1235–1273.
- Rietveld, C. A., van Kippersluis, H., and Thurik, A. R. (2015a). Self-employment and health: Barriers or benefits? Health economics, 24(10):1302–1313.

- Rietveld, C. A., van Kippersluis, H., and Thurik, A. R. (2015b). Self-employment and health: Barriers or benefits? Health Economics, 24(10):1302–1313.
- Schmitz Jr, J. A. (1989). Imitation, entrepreneurship, and long-run growth. Journal of Political Economy, 97(3):721–739.
- Shurchkov, O. and Eckel, C. C. (2018). Gender differences in behavioral traits and labor market outcomes. The Oxford handbook of women and the economy, pages 480–512.
- Skriabikova, O. J., Dohmen, T., and Kriechel, B. (2014). New evidence on the relationship between risk attitudes and self-employment. Labour Economics, 30:176–184. Special Section articles on "What determined the dynamics of labour economics research in the past 25 years? edited by Joop Hartog and and European Association of Labour Economists 25th Annual Conference, Turin, Italy, 19-21 September 2013 Edited by Michele Pellizzari.
- Solon, G., Haider, S. J., and Wooldridge, J. M. (2015). What are we weighting for? Journal of Human Resources, 50(2):301–316.
- Sommers, B. D., Buchmueller, T., Decker, S. L., Carey, C., and Kronick, R. (2013). The affordable care act has led to significant gains in health insurance and access to care for young adults. Health affairs, 32(1):165–174.
- Wellington, A. J. (2001). Health insurance coverage and entrepreneurship. Contemporary Economic Policy, 19(4):465–478.
- Wooldridge, J. M. (1999). Asymptotic properties of weighted M-estimators for variable probability samples. Econometrica, 67(6):1385–1406.
- Yoon, J. and Bernell, S. L. (2013). The effect of self-employment on health, access to care, and health behavior. Health, 2013.
- Yörük, B. K. and Xu, L. (2019). Impact of the ACA's Dependent Coverage Mandate on Health Insurance and Labor Market Outcomes Among Young Adults: Evidence from Regression Discontinuity Design. Eastern Economic Journal, 45(1):58–86.

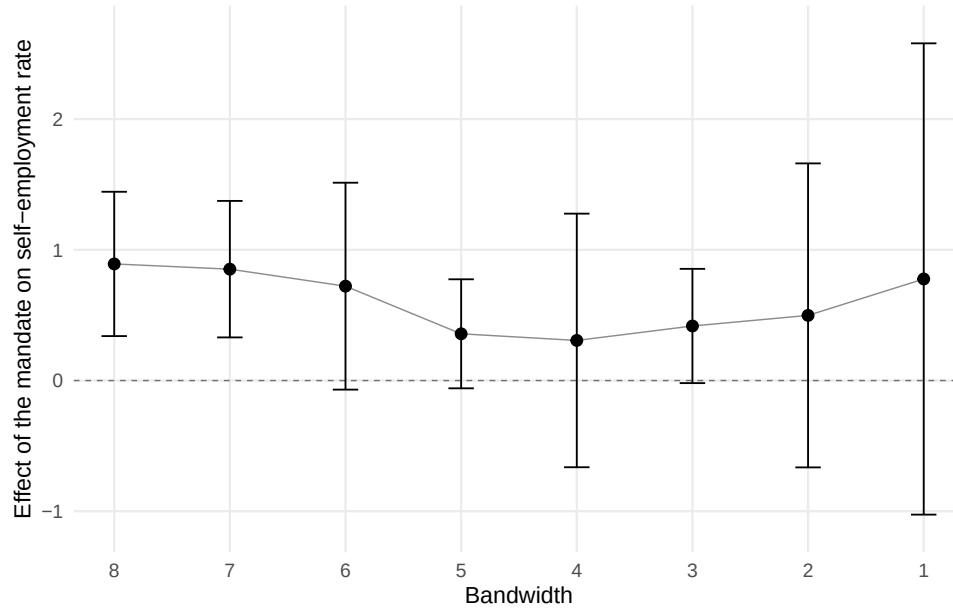
A Robustness Checks

Table A1: Placebo Test

	<i>Outcome Variable:</i>
	Self-employment rate
Placebo Treatment	−0.058 (0.613)
Age FEs	Yes
Year FEs	Yes
Region FEs	Yes
Marital Status FEs	Yes
Race FEs	Yes
Hispanic Ethnicity FEs	Yes
Educational Attainment FEs	Yes
In College FEs	Yes
Family Size FEs	Yes
Observations	49,076
R^2	0.009

Note: This table presents the results of a placebo test using 27–34-year-olds as the treated group and 35–42-year-olds as the control group. Standard errors are in parentheses.

Figure A1: Treatment Effect by Age-Window Bandwidth



Note: Points are difference-in-differences estimates (percentage points). Vertical bars are 90% confidence intervals with standard errors clustered two-way by age and year. Each bandwidth re-estimates the benchmark linear probability model in Table 2. Bandwidth 8 matches the main specification; lower bandwidths narrow the treated and control age windows.

Bandwidth	Treated ages	Control ages
8	19–25 (1), 26 (0.5)	27–34
7	19–25	27–34
6	20–25	27–33
5	21–25	27–32
4	22–25	27–31
3	23–25	27–30
2	24–25	27–29
1	25	27

Table A2: Robustness: Control for Spousal Coverage

<i>Dependent variable:</i>	
Self-employment rate	
Treatment	0.891 ^{***} (0.336)
Spousal coverage	-5.186 ^{**} (1.439)
Fixed effects	Yes
Observations	47,125

Note: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Spousal coverage equals one if the person's health plan is through a spouse's or deceased spouse's previous employer (MEPS establishment type 10), and zero otherwise. Fixed effects match Table 2. Standard errors clustered two-way by age and year are in parentheses.

Table A3: Robustness: Survey-Weighted Baseline

	Unweighted baseline (main text)	Survey-weighted subpopulation
Treatment	0.892*** (0.336)	0.856 (0.597)
Fixed effects	Yes	Yes
Weights / design	None	PERWEIGHT, PSU, strata
Standard errors	Clustered (age, year)	Design-based
<i>p</i> -value for treatment	< 0.01	0.151
Observations	47,125	47,125

Note: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Both columns use the same linear probability specification and fixed effects as Table 2. Coefficients are in percentage points. The survey-weighted column uses `svyglm` on the full MEPS complex survey design with a subpopulation restriction to ages 19–34.

Table A4: Robustness: Logit and Probit Specifications

	<i>Dependent variable:</i>	
	Self-employment rate	
	Logit	Probit
Treatment (AME)	0.647* (0.385)	0.635* (0.373)
Fixed effects	Yes	Yes
Observations	47,125	47,125

Note: *p<0.10; **p<0.05; ***p<0.01. Coefficients are average marginal effects expressed in percentage points. Standard errors clustered two-way by age and year are in parentheses. Fixed effects match Table 2.

B Monte Carlo Study of the Residual-Variance Estimator

This Monte Carlo exercise is not evidence that the MEPS data contain a variance channel. Its narrower purpose is to evaluate whether, when the data-generating process (DGP) contains a latent health-payment variance component σ_i^2 and the researcher observes only one realized annual payment H_i , the squared-residual IV procedure approximates the infeasible oracle under correct specification.

B.1 Design

Each replication draws individuals with MEPS-matched demographic marginals, ages 19–34, and years 2006–2014. The mandate indicator equals one for eligible ages in post-2010 years, with 26-year-olds coded as partially treated.

Health spending is generated as

$$H_i = \mu_i + \sigma_i \varepsilon_i, \quad E[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = 1,$$

where μ_i is the conditional mean of total pay and σ_i^2 is the latent conditional variance. The mandate shifts μ_i and σ_i through principal-component interactions. Realized squared spending satisfies $H_i^2 \neq \sigma_i^2$.

Self-employment depends on the latent moments, not on the realized draw:

$$Y_i = \beta_0 + \beta_1 \mu_i + \beta_v \sigma_i^2 + u_i, \quad u_i = \rho \varepsilon_i + \nu_i,$$

with $\rho = 0.35$ so that OLS on realized H_i and H_i^2 is biased. I compare four estimators: (i) an infeasible oracle that uses true (μ_i, σ_i^2) ; (ii) OLS on realized H_i and H_i^2 ; (iii) an IV fitted-second-moment procedure; and (iv) the empirical squared-residual IV procedure, which instruments $(H - \hat{H})^2$ with mandate-induced variation.

Table B1 summarizes simulation parameters. I consider four DGPs: (i) risk channel only ($\beta_1 = 0, \beta_v < 0$); (ii) subsidy channel only ($\beta_1 \neq 0, \beta_v = 0$); (iii) both channels; and (iv) weak instruments. Each scenario is replicated many times.

Table B1: Monte Carlo Simulation Parameters

Parameter	Value
Observations per replication (paper default)	47,125
Monte Carlo replications (paper default)	500
Diagnostic table below	$N = 8,000$, 100 replications
Endogeneity parameter (ρ)	0.35
Demographic composition	MEPS-matched marginals
Mandate assignment	Ages 19–25 vs. 27–34 (2010–2014)
Outcome DGP	$Y = \beta_0 + \beta_1\mu + \beta_v\sigma^2 + u$
Headline estimator	IV residual variance (Methods section)

Notes: Demographics are drawn from marginal distributions of the analysis sample. Total pay is measured in \$1,000s and mean-centered within each replication. Table B2 reports a diagnostic run; regenerate at paper defaults with `simulation_residual_variance_mc.R`.

B.2 Results

Table B2 reports the risk-channel panel for the variance coefficient β_v . In this diagnostic run, OLS on realized H^2 is badly biased and has the wrong sign. The squared-residual IV estimate has the same sign and similar average bias as the oracle, although its RMSE is larger. The fitted-second-moment IV estimate is also near the oracle in this centered DGP, but its regressor is not itself a variance object. Under the subsidy-only DGP ($\beta_v = 0$), the squared-residual IV estimate remains near zero, while OLS on H^2 falsely rejects a null of no variance effect in essentially every replication.

Table B2: Monte Carlo: Oracle and Feasible Estimates in the Risk-Channel DGP

Estimator	True β_v	Mean $\hat{\beta}_v$	Bias	RMSE	Rej. rate
Oracle variance	-0.0100	-0.0208	-0.0108	0.0847	11.0%
OLS realized H^2	-0.0100	0.0685	0.0785	0.0786	100.0%
IV fitted second moment	-0.0100	-0.0255	-0.0155	0.0913	15.0%
IV squared residual	-0.0100	-0.0244	-0.0144	0.1065	10.0%

Notes: Risk-channel DGP with latent variance σ_i^2 ; the outcome depends on (μ_i, σ_i^2) , not on realized H_i^2 . This diagnostic uses 100 replications with $N = 8,000$. The oracle uses true (μ_i, σ_i^2) . OLS uses realized H_i and H_i^2 . IV second moment uses fitted values from instrumented (H, H^2) . IV squared residual instruments $(H - \hat{H})^2$ with mandate-induced variation. Rejection rate is the share with $|t| > 1.96$.

The simulation evaluates the squared-residual pipeline under stylized DGPs rather than validating it generally. It abstracts from panel dynamics, selective attrition, misspecification of the conditional mean, and measurement error in self-employment. It does not replace the identifying and variance-proxy assumptions required in the field application.